INCH-POUND

A-A-59163B
14 February 2022
SUPERSEDING
A-A-59163A
5 November 2010

COMMERCIAL ITEM DESCRIPTION

# INSULATION TAPE, ELECTRICAL, SELF-ADHERING UNSUPPORTED SILICONE RUBBER

Reactivated after 14 February 2022 and may be used for new and existing designs

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers unsupported silicone rubber insulation tape which will adhere to itself at room temperature. The insulation tape is suitable for use at temperatures from -90 °C to 260 °C.

2. CLASSIFICATION. The insulation tape shall be classified by the following classes and types, as specified (see 7.3(b)):

Class 1 - Tapes conforming to inch-pound dimensions
- Type I - Rectangular cross-section
- Type II - Triangular cross-section with colored guidelines

Class 2 - Tapes conforming to metric dimensions
- Type IM - Rectangular cross-section
- Type IIM - Triangular cross-section with colored guidelines

3. SALIENT CHARACTERISTICS

3.1 First article. When specified (see 7.3(c)), a sample shall be subjected to first article inspection (see 7.5) in accordance with 5.3.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and data which may improve this document should be sent to: DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or emailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil

FSC 5970

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3.2 Material. The insulation tape shall be made of a self-adjusting silicone rubber material which will cure into a homogeneous mass. When wrapped or plied on itself, it will cure to a homogeneous mass at room temperature (21 °C to 27 °C) in 48 hours or in 4 hours at 177 °C.

3.3 Performance and product characteristics. The unsupported silicone rubber material shall have the following performance and product characteristics:

3.3.1 Dimensions and colored guidelines. The insulation tape shall be furnished in the following thicknesses, widths, and colored guidelines as specified in tables I, II, III, and IV.

TABLE I. Type I dimensions.

| | |
|---|---|
| Thickness | 0.010, 0.020, 0.030 ±0.002 inch (in) |
| Width | 0.50, 0.75, 1.00, 1.25, 1.50 inch |

TABLE II. Type II dimensions and colored guidelines.

| Thickness[1] [2] (inch) | Width (inch) | Colored guidelines |
|---|---|---|
| 0.020 | 1.00 | Blue |
| 0.030 | 1.00 | White |
| 0.040 | 1.00 | Green |
| 0.050 | 1.25 | Yellow |
| 0.060 | 1.50 | Blue |
| 0.080 | 1.50 | White |

[1] Thickness tolerance shall be ±0.002 inch.
[2] Thickness edge shall be not less than 0.007 inch and not greater than 0.009 inch.

TABLE III. Type IM dimensions.

| | |
|---|---|
| Thickness | 0.250, 0.500, 0.750 ±0.050 millimeter (mm) |
| Width | 15.0, 20.0, 25.0, 30.0, 40.0 ±0.5 (mm) |

TABLE IV. Type IIM dimensions and colored guidelines.

| Thickness[1] [2] (mm) | Width[3] (mm) | Colored guidelines |
|---|---|---|
| 0.500 | 25.0 | Blue |
| 0.750 | 25.0 | White |
| 1.000 | 25.0 | Green |
| 1.300 | 32.0 | Yellow |
| 1.500 | 38.0 | Blue |
| 2.000 | 38.0 | White |

[1] Thickness tolerance shall be ±0.050 mm.
[2] Thickness edge shall be not less than 0.200 mm and not greater than 0.230 mm.
[3] Width tolerance shall be ±0.5 mm.



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3.3.2 Rolls. Unless otherwise specified (see 7.3(d)), the insulation tape shall be supplied in rolls of specified thickness and width (see 3.3.1) and wrapped on a core with insulation tape interleaved with quick-release film. The quick-release film shall be of sufficient width to preclude insulation tape edges from fusing during packaging, packing, and storage. Unless otherwise specified (see 7.3(d)), the insulation tape shall be supplied in rolls not less than 11 meters (12 yards) in length.

3.4 Physical properties. The physical properties of the unsupported silicone rubber material shall be as specified in table V.

TABLE V. Physical properties.

| Property | Value types I and II | Value types IM and IIM | Test method |
|---|---|---|---|
| Tensile strength,<br>pound force per square inch ($lbf/in^2$)<br>kilopascal (kPa) | 700 minimum | 4,800 minimum | ASTM D3759/ D3759M |
| Elongation,<br>percent | 300 minimum | 300 minimum | ASTM D3759/ D3759M |
| Water absorption[1],<br>percent by weight | 3 maximum | 3 maximum | FED-STD-601, method 6251 |
| Volume resistivity,<br>ohms/circular mil/foot<br>nanoohm meter | $1 \times 10^{13}$ | $1.7 \times 10^{13}$ | ASTM D257 |
| Bond strength,<br>1 inch width, pounds<br>25 mm width, kilogram-force (kgf) | 2.0 minimum | 0.9 minimum | ASTM D2148 |
| Dielectric strength,<br>volts/mil[2] tape thickness | | | ASTM D149[3]<br><br>ASTM D2148[4] |
| 0.010 in 0.250 mm | 400 minimum | 400 minimum | |
| 0.020 in 0.500 mm | 400 minimum | 400 minimum | |
| 0.030 in 0.750 mm | 400 minimum | 400 minimum | |
| 0.040 in 1.000 mm | 300 minimum | 300 minimum | |
| 0.050 in 1.300 mm | 275 minimum | 275 minimum | |
| 0.060 in 1.500 mm | 250 minimum | 250 minimum | |
| 0.080 in 2.000 mm | 200 minimum | 200 minimum | |

[1] See 3.5 for preparation of test specimen.
[2] This requirement is expressed in volts per thousandths inch thicken+ss for both metric and inch-pound products.
[3] For types I and IM.
[4] For types II and IIM.



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3.5 Preparation of test specimen for water absorption test. For types II and IIM, three strips of insulation tape shall be lapped together as shown in figure 1 to provide material for a dumbbell shaped die cut test specimen with uniform thickness.

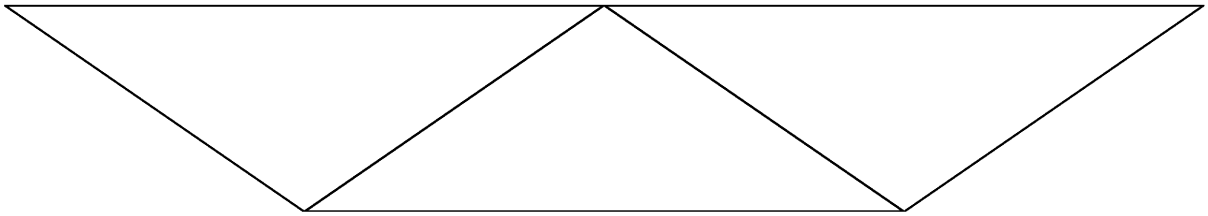


FIGURE 1. Preparation of insulation tape specimen for water absorption test.

3.6 Storage life. Insulation tape stored in unopened containers at a temperature of not greater than 37.3 °C shall be capable of meeting all of the requirements of this specification for a period of 12 months from date of manufacture. (For storage life extension, see 7.4.)

3.7 Workmanship. The insulation tape shall be uniform in quality and shall be free from splits, cracks, foreign material, and other defects that would affect life, serviceability, or appearance.

## 4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

## 5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The products offered must have been previously sold either to the government or on the commercial market.

5.3 First article. When specified (see 7.3(c)), a sample shall be subjected to first article inspection in accordance with the tests specified in section 3.

## 6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(e)).



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7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

AA59163 - X XXX XXXX

| | | |
|---|---|---|
| Thickness | 0020 | 0.020 in |
| | 0030 | 0.030 in |
| | 0040 | 0.040 in |
| | 0050 | 0.050 in |
| | 0060 | 0.060 in |
| | 0080 | 0.080 in |
| | 0250 | 0.250 mm |
| | 0500 | 0.500 mm |
| | 0750 | 0.750 mm |
| | 1000 | 1.000 mm |
| | 1300 | 1.300 mm |
| | 1500 | 1.500 mm |
| | 2000 | 2.000 mm |
| Type | I | Rectangular cross-section |
| | II | Triangular cross-section with colored guidelines |
| | IM | Rectangular cross-section |
| | IIM | Triangular cross-section with colored guidelines |
| Class | 1 | Tape to inch-pound dimensions |
| | 2 | Tape to metric dimensions |
| CID number | | |

Examples of reference part numbers: inch-pound AA59163-1I0020
metric AA59163-2IIM0250

7.2 Sources of documents.

7.2.1 Federal standards. Copies of federal standards may be obtained from General Services Administration, Federal Supply Service, Specification Section, 470 East L'Enfant Plaza SW, Suite 8100, Washington, DC 20407. Electronic copies may be obtained from https://assist.dla.mil.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from the ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies may be obtained from https://www.astm.org/.



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7.2.3 Federal Acquisition Regulation (FAR). Electronic copies may be obtained from https://www.acquisition.gov/browse/index/far.

7.3 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Class and type (see 2.).
c. First article (see 3.1 and 5.3).
d. When width, length, or thickness is other than specified (see 3.3.2).
e. Preservation, packing, and marking (see 6.1).

7.4 Storage life extension. Insulation tape that has exceeded storage life may be examined for compliance with table V. Acceptable test results shall validate storage life extension for an additional 6 months (see 3.6).

7.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a number of items. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bid should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending contract.

7.6 Subject term (key word) listing.

Bond
Cure
Dielectric
Lapped
Plied
Polymer
Roll
Self-adjusting
Wrapped



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MILITARY INTERESTS:

Custodians:
Army - MI
Air Force - 85
DLA - GS

Review Activity:
Air Force - 99

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:
DLA – GS6
(Project 5970-2022-001)

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